

EXPERIMENT 25 : FREQUENT PATTERN MINING USING ASSOCIATION RULE THROUGH WEKA AND R -TOOLS

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associator

Choose **Apriori** -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1

Start Stop

Result list (right-click for ...)

19:29:11 - Apriori

19:31:11 - Apriori

Associator output

```
==== ASSOCIATOR MODEL (full training set) ====

Apriori
=====

Minimum support: 0.15 (694 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 17

Generated sets of large itemsets:

Size of set of large itemsets L(1): 44
Size of set of large itemsets L(2): 380
Size of set of large itemsets L(3): 910
Size of set of large itemsets L(4): 633
Size of set of large itemsets L(5): 105
Size of set of large itemsets L(6): 1

Best rules found:

1. biscuits=t frozen foods=t fruit=t total=high 7
2. baking needs=t biscuits=t fruit=t total=high 7
3. baking needs=t frozen foods=t fruit=t total=high 7
4. biscuits=t fruit=t vegetables=t total=high 81
5. party snack foods=t fruit=t total=high 854 ==
6. biscuits=t frozen foods=t vegetables=t total=
7. baking needs=t biscuits=t vegetables=t total=
8. biscuits=t fruit=t total=high 954 ==> bread a
9. frozen foods=t fruit=t vegetables=t total=high
10. frozen foods=t fruit=t total=high 969 ==> bre
```

weka.gui.GenericObjectEditor

weka.associations.Apriori

About

Class implementing an Apriori-type algorithm.

car False

classIndex -1

delta 0.05

doNotCheckCapabilities False

lowerBoundMinSupport 0.1

metricType Confidence

minMetric 0.9

numRules 10

outputItemSets False

removeAllMissingCols False

significanceLevel -1.0

treatZeroAsMissing False

upperBoundMinSupport 1.0

verbose False

Open... Save... OK Cancel

[155] convr: (3.35)
[149] convr: (3.28)
[103] [150] convr: (3.27)
[159] convr: (3.26)
convr: (3.15)
[0.03] [151] convr: (3.06)
[0.03] [145] convr: (3.01)
[3] [156] convr: (3)
[2.92]

Status

OK

Log

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associator

Choose **Apriori** -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1

Start Stop

Result list (right-click for ...)

19:29:11 - Apriori

Associator output

```
[List of attributes omitted]
==== Associator model (full training set) ====

Apriori
=====

Minimum support: 0.15 (694 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 17

Generated sets of large itemsets:

Size of set of large itemsets L(1): 44
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Size of set of large itemsets L(5): 105
Size of set of large itemsets L(6): 1

Best rules found:

1. biscuits=t frozen foods=t fruit=t total=high 788 ==> bread and cake=t 723 <conf:(0.92)> lift:(1.27) lev:(0.03) [155] convr:(3.35)
2. baking needs=t biscuits=t fruit=t total=high 760 ==> bread and cake=t 696 <conf:(0.92)> lift:(1.27) lev:(0.03) [149] convr:(3.28)
3. baking needs=t frozen foods=t fruit=t total=high 770 ==> bread and cake=t 705 <conf:(0.92)> lift:(1.27) lev:(0.03) [150] convr:(3.27)
4. biscuits=t fruit=t vegetables=t total=high 815 ==> bread and cake=t 746 <conf:(0.92)> lift:(1.27) lev:(0.03) [159] convr:(3.26)
5. party snack foods=t fruit=t total=high 854 ==> bread and cake=t 779 <conf:(0.91)> lift:(1.27) lev:(0.04) [164] convr:(3.15)
6. biscuits=t frozen foods=t vegetables=t total=high 797 ==> bread and cake=t 725 <conf:(0.91)> lift:(1.26) lev:(0.03) [151] convr:(3.06)
7. baking needs=t biscuits=t vegetables=t total=high 772 ==> bread and cake=t 701 <conf:(0.91)> lift:(1.26) lev:(0.03) [145] convr:(3.01)
8. biscuits=t fruit=t total=high 954 ==> bread and cake=t 866 <conf:(0.91)> lift:(1.26) lev:(0.04) [179] convr:(3)
9. frozen foods=t fruit=t vegetables=t total=high 834 ==> bread and cake=t 757 <conf:(0.91)> lift:(1.26) lev:(0.03) [156] convr:(3)
10. frozen foods=t fruit=t total=high 969 ==> bread and cake=t 877 <conf:(0.91)> lift:(1.26) lev:(0.04) [179] convr:(2.92)
```

Status

OK

Log